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Navigating the shift from digital to AI literacy: reimagining the computer science curriculum for human-AI collaboration in Zimbabwean higher education

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ABSTRACT

Computer Science (CS) curricula require urgent redesign as generative AI automates tasks once performed by programmers. This study examined how CS curricula can be reimagined to remain relevant by shifting from traditional digital literacy toward AI literacy and preparation for human-AI collaborative work. Guided by a framework combining the Technology Acceptance Model, Diffusion of Innovations Theory and Constructivist Learning Theory, the study explored factors shaping curriculum transformation, AI adoption dynamics and the social construction of AI literacy. Using a qualitative interpretive design, nineteen purposively selected participants - CS students, ICT managers, and university lecturers were interviewed through semi-structured interviews, with data analysed thematically to identify recurring patterns. Four major themes emerged: the evolving definition of "literacy" in the AI era; curriculum gaps marked by overemphasis on traditional programming at the expense of AI literacy and critical thinking; barriers to AI-oriented reform, including lecturer preparedness, curriculum overload, and infrastructural constraints; and strategies for transformation centred on industry-academia collaboration, project-based learning, interdisciplinary integration, AI ethics, and African-centred datasets for applied problem-solving. The study proposes a resource-aware AI literacy framework for reimagining CS curricula. Findings suggest that transitioning toward AI literacy demands collaborative, context-sensitive, and ethically grounded reform to prepare graduates for AI-mediated professional environments.

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1. Introduction

The rapid pace of technological development in the twenty-first century has a significant impact on the way people interact with information, one another, and the external world (Yi, 2021). Digital literacy has been a major player in this change, serving as the foundation of modern education and workforce preparedness for over a century. Digital literacy includes being able to navigate the digital world effectively; that is, to access, manage, integrate, evaluate, create, and communicate digital information using any technology (Kaur, 2024; Rustemi & Tahiri, 2024). However, this skill set is being supplemented with AI literacy and, even to a certain extent, supplanted (Tiernan et al., 2023). Existing digital literacy frameworks have been very slow in acknowledging AI advancements (Ismaili, 2024).

AI in education refers to the use of computer systems that mimic human intelligence to improve teaching and learning (Ndlovu et al., 2025; Chimbo et al., 2026). As AI technology transitions from a specialised area to a core technology, the requirements for digital competencies are undergoing a metamorphosis from basic digital literacy to an in-depth understanding of AI concepts, capabilities, ethical considerations, and practical applications (Amit Dubey et al., 2024; Moyo et al., 2024). The introduction

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of highly advanced generative AI tools across industries has caused a significant shift, requiring a reconsideration of the definition of digital literacy and the skills needed for the future workforce (Amit Dubey et al., 2024).

Curricula in educational institutions, especially in Computer Science (CS), will have to face the challenge head-on, in which students are not only taught how to use such technologies but also why and where to use them (Wimmer et al., 2024). Unlike past cases where technology only enhanced human capabilities, generative AI today can perform tasks once thought to be within the domain of human programmers, not only coding and producing human-like text, but also solving complex problems (Bengesi et al., 2024).

The CS curriculum primarily focuses on operating systems, core programming, algorithms, data structures, and technical skills from the pre-AI digital era. It is now an urgent necessity to evolve and equip students for the AI-driven environment (Walter, 2024; Zhang, 2025). In addition, the tightening job market for technology roles exacerbates the issue, as graduates face greater competition for positions requiring AI-related skills; thus, the need for change is becoming clearer (Bengesi et al., 2024). While in the past graduates had no problem finding jobs, they now face intense competition, as many entry-level programming roles have been taken over by AI, making it more difficult for CS graduates to secure a job without specialised, AI-relevant skills (Zhang, 2025). Industry leaders and experts are becoming increasingly vocal about their speculations regarding AI's capability to match that of mid-level software engineers, thus creating considerable urgency within the technology sector and, consequently, academia as well (Mastropaolo et al., 2024; Picciano, 2019). This technological evolution necessitates an upgrade in skills from basic digital literacy to more discerning, comprehensive AI literacy within the CS field (Tiernan et al., 2023; Baskara, 2025). Even though it is widely accepted that AI literacy should be part of the curriculum (Baskara, 2025; Zhang et al., 2025), many CS schools still focus more on traditional technical skills and programming languages (Zhang et al., 2025). The gap between curricula and industry needs results in poor student readiness for the new labour market where AI technologies are dominant (Radermacher & Walia, 2013; Kabashkin et al., 2023).

There is consensus on the necessity of AI integration in CS education and the urgency of redesigning the curriculum to produce graduates for an AI-intensive workforce. However, a significant gap and complex challenge remain in understanding the diverse perspectives of key stakeholders (Zhang et al., 2025; Eden et al., 2024). There is a limited understanding of how the main stakeholders, who are either involved in or affected by CS education, perceive this transition and the practical challenges and opportunities that accompany it.

This qualitative research aims to fill this gap by identifying diverse stakeholder views and experiences regarding the transition from digital to AI literacy and the rethinking of the CS curriculum, thereby contributing to the development of strong, future-proof CS education strategies. Moreover, it will explore how educational frameworks could be reimagined to meet new demands, so that graduates are not only adequately but also confidently prepared for the challenges of the AI era. If stakeholders' perspectives are not well understood, the future-proof CS curriculum will not be aligned with changing industry needs, educational feasibility, and student preparedness, potentially leaving graduates unprepared for the rapidly evolving AI-driven professional landscape.

2. Literature review

This section reviews the evolving conceptualisation of digital literacy in the context of rapid advancements in AI. It situates AI literacy within broader debates on technology-enhanced education and examines how traditional understandings of digital competence are being redefined. The review further establishes the theoretical and conceptual foundations that inform this study, with particular attention to the transition from general digital skills towards more complex, context-aware AI capabilities in higher education.

2.1. Reconceptualising digital literacy for AI literacy

AI literacy and digital literacy are distinct concepts, yet they are often conflated in educational discussions. Digital literacy is a skill that includes finding, evaluating, creating, and communicating information through digital means (Kaur, 2024; Rustemi & Tahiri, 2024). It encompasses the entire spectrum of skills, ranging from the very basic, such as using a computer and the internet, to advanced ones, including online information and digital communication (Kaur, 2024; Vodă et al., 2022). On the other hand, AI literacy encompasses technology skills, knowledge of the principles, limitations, data, and skills required to work with AI, and the capacity to think critically about AI's role in society and ethics (Kaur, 2024; Rustemi & Tahiri, 2024). Emerging scholarship increasingly positions AI literacy as part of a broader notion of "AI fluency," which extends beyond basic interaction with AI systems towards adaptive, critical, and context-aware engagement with intelligent technologies across professional and societal settings (Rogers & Carbonaro, 2025).

In this research, AI literacy is grounded in digital literacy, which facilitates the transition from using digital tools to understanding and interacting with intelligent systems. Computer science graduates will be expected to be proficient in programming languages and AI use in their sector, and to comprehend responsible AI development and deployment (Borenstein and Howard, 2021). Worldwide educational systems have justifiably made investments in this vital skill set (Borenstein & Howard, 2021). However, literacy is quickly changing its boundaries. The use of AI technologies is widespread, and even simple ones, like generative models that automate coding and analysis, have made technological proficiency still a basic skill. This basic conceptual shift from "technological proficiency" to "cognitive and ethical mastery" is the philosophical driving force behind the worldwide demand for reforming CS curricula. Additionally (Lamberti, 2024), notes that many educational institutions lack the proper framework to incorporate AI principles into their curricula. This gap constitutes a significant challenge for lecturers, who must prepare students for a future in which AI is likely to be integrated into almost every aspect of life.

2.2. Curriculum architecture: the challenge of balancing rigour and relevance

The world has acknowledged the need to rethink the CS curriculum by incorporating AI (Zhang, 2025; Eaton & Epstein, 2024). The inevitability of an AI-imbibed environment has forced universities all over the globe to redo their CS programs and restructure curricula (Southworth et al., 2023). The call to integrate AI principles has created a significant architectural challenge: curriculum overload. The traditional CS school has a sound theoretical basis. However, universities must find a solution that integrates the expansive scope of modern AI (e.g. machine learning, deep learning, and ethics) without sacrificing the essential foundational theoretical base of CS (e.g. algorithms and complexity theory) (Hutson and Ceballos, 2023). First, the solution is integration, not addition. The most effective approach moves away from ad hoc specialisation and advocates transdisciplinary pathways that fuse technical training with humanities and social science skills (Hutson & Ceballos, 2023), through multidisciplinary summer schools (Mackenzie et al., 2024) and the ethical enhancement of AI using quantum computing (Olorunsogo et al., 2024). This requires teaching foundational principles through the context of applied AI problems. Second, the solution requires a pedagogical mandate. AI requires skills such as ethical judgment and iterative problem-solving; it cannot be effectively taught through traditional, descriptive lectures (Saltz et al., 2019). Other research promotes modifying curricula based on institutional identity and a liberal arts perspective, in contrast to ad hoc composite approaches (Holland-Minkley et al., 2023).

2.3. The global shift in re-imagining the CS curriculum: the contextual and theoretical gap

Prominent CS institutions, such as the University of Florida, are implementing AI across the curriculum to equip all students with AI knowledge and opportunities, thereby cultivating an AI-ready workforce (Southworth et al., 2023). The University of Cape Town, located in South Africa, has faculty members who are actively participating in conversations about the changes in their programs, knowing that the basics of computing are nevertheless very important, but also a drastic change in curriculum structure and content is needed (Collier-Reed & Committee, 2025). In a constantly changing world, AI literacy

frameworks are emerging, requiring a curriculum that is always up to date and flexible. Some researchers argue that it is necessary to create a framework for integrating AI into CS education, with a primary focus on basic AI knowledge, practical applications, and discussions of ethical and societal issues (Zhang, 2025; Eaton & Epstein, 2024).

Similarly, while international scholarship has emphasised cognitive and ethical competences for AI literacy (Patra, 2025), in the context of African studies, the focus has been on structural disparities, infrastructural deficiencies, a lack of locally relevant, culturally grounded case studies, and policy deficiencies. Scholarship on the continent indicates that the situation is no longer about adopting AI or training graduates in AI, but rather about the urgency of getting graduates involved in creating localised AI solutions. These graduates should be data colonialism-free, i.e. free of AI models applied without ethical consideration for local sovereignty or socio-economic realities (Eden et al., 2024; Patra, 2025). This comparison between international scholarship and African studies implies that, in Africa, the challenge is not a shortfall in understanding the concepts but rather the application of AI literacy principles in curriculum development, which is hindered by resource constraints. The synthesis indicates three interrelated streams of scholarship: (1) the theoretical conceptualisation of AI literacy as a competency for the digital era; (2) pedagogical frameworks connecting AI integration to constructivist educational practices; and (3) contextual evaluations of AI policy preparedness within African higher education. These strands converge to reveal a persistent gap: the lack of frameworks that combine core CS rigour with AI-driven innovation. Moreover, existing frameworks and instructional guidelines are mostly theoretical or review-based, lacking strong empirical grounds for their effectiveness (Baskara, 2025; Casal-Otero et al., 2023).

By shifting the conventional emphasis from basic programming skills to hybrid courses, where computing is combined with other disciplines, lecturers can foster a systemic understanding of technology that will empower current generations to thrive in the ever-changing digital world (Zhang, 2025; Southworth et al., 2023). However, it is acknowledged that there will be difficulties in rolling out an AI-driven curriculum, including faculty training, resource allocation, overcrowding, and the rapid pace of change in AI technologies. Plus, the challenge of bridging the gap between graduates' skills and industry expectations, and of addressing ethical issues such as data privacy and potential biases, are significant concerns that must be addressed in the AI integration process (Eden et al., 2024; Ndlovu & Maguraushe, 2025).

The complexity of this reform, which involves institutional pressures, faculty attitudes, and pedagogical methods, calls for an integrated framework discussed in the next section.

3. Theoretical underpinning

Three critical theories, the Diffusion of Innovations (DOI) Theory, the Technology Acceptance Model (TAM), and the Constructivist Learning Theory, offer important insights into how CS curricula can be modified to meet the requirements of AI literacy.

3.1. Diffusion of innovations (DOI) theory

The DOI Theory (Rogers et al., 2019) explains how novel ideas and technologies diffuse within a society, including the rationale and speed of diffusion. The theory stresses the role of communication channels or systems and the perceived attributes of innovations on acceptance. The five attributes identified by Rogers which are pertinent to this study in designing efforts to take AI literacy into the curriculum are: relative advantage: The advantages of implementing AI literacy and an updated curriculum over existing approaches; compatibility: To what extent does the transition to AI literacy and the updated curriculum align with existing educational philosophies, institutional frameworks, and industry requirements?; complexity: The apparent complexity of teaching AI concepts, designing new courses, and integrating AI literacy competencies into the curriculum can hinder its acceptance; trialability: The degree to which novel AI-centric modules or pedagogical approaches can be tested on a restricted scale before comprehensive implementation, and observability: The degree to which the outcomes of an innovation are perceptible

to others. This study defines "innovation" as a shift from standard digital literacy to AI literacy, thereby "reimagining" the CS curriculum.

3.2. Technology acceptance model (TAM)

TAM came about as a theory to explain how users accept and use new technologies (Davis, 1989). The model identifies two major factors affecting the acceptance of any technology: perceived usefulness and ease of use. Perceived usefulness can be evaluated by assessing whether AI tools assist lecturers in teaching and learning, which, in turn, correlates with the production of better-trained students. In the industry, this leads to more efficient hiring. From a student's perspective, it also offers better job prospects. From an instructor's perspective, perceived ease of use refers to the extent to which using AI technology requires little effort (Davis, 1989), which in turn affects the incorporation of these technologies into their teaching methods, the curriculum designers' willingness to implement changes, and students' ability to learn new AI-related material. Understanding these characteristics is crucial for curriculum development that imparts AI concepts and fosters the adoption of AI tools in educational settings.

3.3. Constructivist learning theory (CLT)

Constructivist Learning Theory (CLT) (Schwebel, 1979) holds that individuals learn and acquire knowledge primarily through interaction with their surroundings and peers. The theory strongly emphasises the importance of learners' active participation, teamwork, and critical thinking in the educational process. Therefore, learners build knowledge actively as they strive to understand their environment. Such a viewpoint is not only necessary to understand how different people independently interpret "AI literacy" or "the reimagined curriculum" based on their different roles and experiences, but also reflects on the issue of personal understanding being socially negotiated and improved through interaction, thus leading to sharing or diverging views and methods in the curriculum development process. The understanding that AI literacy will entail a long learning curve for lecturers, in terms of new content and pedagogical approaches, and for students, in terms of new concepts, is crucial. These learning paths will significantly affect the acceptance and execution of change. Constructivism concerning AI literacy is ready to provide support to lecturers in professional development as it helps them to redefine their pedagogical practices and teaching environments where students are involved in the AI technology interactions for the sake of teaching the implications and functions of the technology and at the same time, graduates are not just passive programmers, but active co-creators who can adapt to the constant evolution of AI (Richter et al., 2025).

3.4. Integrated framework and theoretical gap

This study is based on a multi-theoretical framework. It will examine the transition from digital to AI literacy and the reform of the CS curriculum. The TAM and DOI theories both focus on technology adoption, but when combined with CLT, they form a powerful, comprehensive framework. TAM takes into account users' individual opinions about the technology's usefulness and the effort required to use it, thereby indirectly influencing the acceptance of the associated ideas and changes. DOI, on the other hand, sheds light on the institutional and systemic characteristics, such as relative advantage, complexity, and organisational readiness, that are necessary for curriculum innovation. These two theories together provide a comprehensive understanding of the behavioural, pedagogical, and structural dimensions of AI adoption processes. CLT links the various perspectives on adoption by emphasising that learning occurs after the technology is already in place and that lecturers must change their practices to sustain AI literacy. It presents AI literacy as an active, social, and experiential process, aligning with the requirement for collaborative, project-based learning.

In higher education institutes in Africa, notably in Zimbabwe's Education 5.0 framework (Mupaikwa, 2014), such a multi-theoretical approach is critical. The TAM and the DOI delineate the factors that drive AI integration. In contrast, CLT situates this transition within culturally responsive, student-centred

pedagogy. By doing so, they construct an empirically tested framework that is conceptually consistent for the simultaneous critique of the acceptance and the deep-seated integration of AI literacy into the CS curricula. Such a union of disciplines enables research to progress from simple adoption to a deeper understanding of pedagogical change, thereby demonstrating that disseminated innovation is not only a new syllabus but also a new educational philosophy. No existing framework is known to the researcher that analytically combines these three critical features to produce an empirically grounded, practical model for sustainable reform in resource-constrained situations.

4. Methodology

This study employed a qualitative, interpretive research approach. Purposive sampling was used to select participants with a wealth of information relevant to the research questions (Patton, 2015).

4.1. Recruitment and sampling procedure

ICT Managers: Industry experts from the ICT sector who oversee internships and practical training for CS students. They provide insights into the skills needed in the workforce, which is vital for understanding industry needs, the skills gap, and the practical application of AI literacy in the workplace. To be included, they needed more than 10 years of industry experience.

Lecturers: Individuals actively involved in teaching CS courses and/or designing CS curricula at higher education institutions. Their viewpoints are essential to understanding curriculum integration, pedagogical challenges, institutional preparedness, and the systemic opportunities and obstacles in curriculum reform. To be included, they needed more than 10 years of academic experience.

Students: These are either pursuing advanced education, such as postgraduate studies, or are almost finished, i.e. in their fourth year with their CS degrees. Their experiences with the current curriculum and their opinions of the demands of future careers provide a distinct "user" perspective.

To recruit lecturers, the researcher contacted the program coordinators of CS departments at four state universities in Zimbabwe which offer a CS degree. For confidentiality purposes, participating universities are anonymised and referred to as Universities A-D throughout the study. Focused outreach to student organisations interested in technology and AI was used to identify students. Focused searches on professional networking sites such as LinkedIn were used to identify industry professionals. Using a snowball sampling technique, participants were also asked to recommend additional eligible individuals for the study. A total of 19 participants were included.

Basic demographic information, including participants' professional roles and years of experience in their field/institution, was collected to provide context (Table 1). This data helped contextualise their perspectives without revealing individual identities.

4.2. Data collection method and interview protocol design

Semi-structured interviews were conducted to facilitate guided discussion and to allow for in-depth exploration of emergent themes (Salhab, 2024). The interview protocol was designed to elicit rich qualitative data on participants' understanding of AI literacy, their views on the current CS curriculum, perceived challenges and opportunities in integrating AI, and their vision for future curriculum design. Table 2 shows how each interview question was theoretically aligned to elicit perceptions corresponding to the constructs of the three frameworks. This alignment ensured that the interviews captured cognitive, institutional, and pedagogical dimensions shaping the shift from digital to AI literacy.

Interviews were conducted through Microsoft Teams, and each session lasted approximately 40–60 minutes, allowing for thorough conversations. An information sheet explaining the study's purpose, participants' rights, and the confidentiality guarantee was sent to all participants before the interview. Each participant gave verbal and written informed consent before the interview began. For the remote sessions, the built-in recording feature of Microsoft Teams was used, and the automated transcripts served as the starting point. They were then transcribed word-for-word by the researcher. The researcher

Table 1. Demographic details of participants.

Designation	Gender	Years of experience in the respective field	University
ICT Manager1	Male	10+	
ICT Manager2	Male	10+	
ICT Manager3	Female	10+	
ICT Manager4	Male	15+	
ICT Manager5	Male	10+	
ICT Manager6	Male	10+	
Lecturer1	Female	10+	A
Lecturer2	Male	15+	A
Lecturer3	Male	10+	B
Lecturer4	Male	10+	B
Lecturer5	Male	10+	C
Lecturer6	Male	20+	D
Lecturer7	Female	10+	D
Lecturer8	Female	10+	D
Student1	Male		A
Student2	Female		D
Student3	Female		C
Student4	Male		D
Student5	Male		B

Table 2. Guiding data collection: Theoretical frameworks mapped to interview protocol.

Theoretical framework	Core construct focus	Aligned interview questions
TAM *These questions explored participants' perceptions of the usefulness and ease of implementing AI-integrated teaching approaches.*	Perceived Usefulness (PU) & Perceived Ease of Use (PEOU)	Q1. What are the key strengths and weaknesses of the current Computer Science curriculum in preparing students for the AI era? Q6. What challenges do you foresee in shifting from traditional digital literacy to AI literacy within the curriculum?
DOI *These questions probed how participants perceive the advantages, challenges, and institutional dynamics influencing AI curriculum diffusion.*	Relative Advantage, Complexity & Trialability	Q4. How do you envision the "reimagined" Computer Science curriculum that integrates AI literacy to look like in the next 5–10 years? Q5. What role should industry and universities play in driving this curricular change? Q7. Can you share examples of successful AI literacy initiatives you have encountered?
CLT *These questions elicited insights on collaborative, experiential, and project-based learning as key elements of developing AI literacy.*	Social Construction of Knowledge & Experiential Learning	Q2. How do you define AI literacy in the context of a Computer Science graduate? Q3. What essential AI-related competencies should every Computer Science graduate possess?

diligently compared each transcript with the original audio to ensure complete accuracy and to include nonverbal cues and pauses where necessary.

4.3. Data analysis

Atlas.ti software was used for analysis. Thematic analysis was employed to identify, analyse, and report patterns within the qualitative data (Braun & Clarke, 2006). The coding technique used an inductive-deductive approach to ensure that both data and theory are fully accounted for in the interpretation. The researcher reviewed the transcribed interviews several times to become familiar with the data. After that, open coding was conducted in Atlas.ti. The initial codes were thus developed inductively, meaning they did not rely on any predetermined categories. Rather, they allowed the participants' voices to lead them to the first findings. By comparing and refining codes, similar ones were grouped into larger clusters, revealing underlying patterns.

In continuation, the axial coding was performed to relate these categories to the TAM, DOI, and CLT theories. It linked the participant's experiences to theoretical constructs such as perceived usefulness, innovation complexity, and social knowledge construction. The process ensured that the DOI, TAM, and CLT frameworks contributed to the conceptualisation of the new themes, making the analysis more

Table 3. Intercoder agreement summary.

Coding aspect	Description
Coding approach	Independent double-coding of selected transcripts
Reliability measure	Percentage agreement
Agreement level	88%
Resolution process	Discussion and consensus refinement of codes
Final outcome	Harmonised coding framework and validated themes

coherent and in-depth. The researcher then validated the suggested themes against the entire dataset, confirming internal homogeneity (within themes) and external heterogeneity (between themes). Minor themes that lacked sufficient supporting evidence were either merged or dropped.

A final round of selective coding grouped the main themes that best represented the transition from digital literacy to AI literacy. To enhance coding trustworthiness, the study employed a qualitative data analyst to conduct an independent evaluation of the transcripts. To enhance coding trustworthiness, an independent qualitative data analyst coded a subset of the transcripts separately from the primary researcher. Intercoder agreement was assessed using percentage agreement, which yielded an agreement level of 88% across the independently coded segments. Coding discrepancies were subsequently reviewed through discussion and resolved through consensus, resulting in refinement and alignment of the coding framework. Throughout this process, reflexive memos were kept documenting the analytical decisions and to clarify the procedure. Table 3 presents the intercoder agreement summary.

In this qualitative study, data saturation served as the criterion for evaluating sample adequacy. Based on the methodology proposed by (Salhab, 2024), recruitment and interviews for each group were conducted sequentially and stopped when no further significant themes or insights related to the research topics were identified. After the 16th interview, responses began to show considerable overlap in topics, and the last three interviews confirmed that no new insights were emerging. As the sample comprised different stakeholder groups (8 lecturers, 5 students, and 6 ICT managers), this consistency across categories indicated topic saturation rather than mere repetition within a single group. Though the study does not claim statistical generalizability, achieving saturation ensured conceptual depth and interpretive validity, in line with qualitative research standards.

4.4. Credibility and trustworthiness of findings

Semi-structured interviews were the primary source of data. However, this research also employed methodological triangulation to enhance credibility and interpretive validity. Apart from getting the firsthand views, a documentary analysis of relevant institutional and policy materials was done, which included course synopsis, departmental course outlines, university strategic plans, and national frameworks like Heritage-based Education 5.0 (Mupaikwa, 2014), the Ubuntu philosophy (Waghid, 2020; Mangena, 2019), the National Artificial Intelligence Strategy (2026–2030) (MICTPCS\) Ministry of ICT, 2026) and the African Union Continental Strategy for Education (CESA 2016–2025) (African Union, 2015; Awaah, 2024). These sources helped to cross-validate participant claims and verify the alignment (or mismatch) between perceived institutional readiness and documented curricula. This triangulated method, along with the theoretical triangulation through DOI, TAM, and CLT, enhanced the study's methodological rigour, ensuring that the interpretations were both empirically grounded and contextually authentic

4.5. Ethical considerations

The study adhered to strict ethical guidelines to protect the rights and well-being of all participants. Ethical approval was obtained from the Ethical approval was obtained from the National University of Science and Technology Institutional Review Board (NUST-IRB) (Approval Ref: NUST/IRB/2024/074) before any data collection commenced. All participants received comprehensive information about the study's purpose and procedures before voluntarily participating and provided their informed consent. Participant anonymity was ensured by assigning pseudonyms and removing identifying information

from transcripts and final reports. All data were stored securely. Participants were informed of their right to withdraw from the study without penalty.

5. Results

Four main themes emerged, illustrating how participants felt about and experienced the transition from digital to AI literacy and the re-evaluation of the CS curriculum. It is important to note that the views expressed in participant quotations reflect individual perceptions and experiences and should not be interpreted as evaluative judgments by the authors

5.1. Theme 1: the evolving definition of 'literacy' in the AI era

Lecturers pointed out that they were placing increasing emphasis on computational thinking and engagement with AI technology, rather than focusing on rote programming skills. They also clearly distinguished everyday digital literacy from the new notion of AI literacy, which requires a deeper understanding than simply using the tool. Digital literacy was defined as the basic skill of using digital technology, for example, Microsoft Word. On the other hand, AI literacy means understanding basic algorithms and how they make decisions, being aware of the ethical implications, and critically engaging with AI systems. It also encompasses detecting bias and recognising the situations where AI should not be used. Lecturers revealed that a "liberal arts approach" is required in CS, meaning combining technical expertise with concepts such as data bias, model learning, and real-world effects.

Lecturer3 *"The future of computer science education may increasingly resemble a liberal arts approach, where critical thinking and effective communication are valued alongside technical skills. This shift reflects a broader understanding of what it means to be literate in an AI-driven world."*

Lecturer4 *"AI literacy involves understanding and utilising AI technologies. We need graduates who thoroughly understand AI, its limitations, the data sources, and the ethical ramifications. Implementing a neural network responsibly is just as important as programming it. For us, that is true AI literacy"*.

Lecturer6 *"Capacity to engage, create, implement, and responsibly oversee AI systems." Digital literacy means using programs like Word, but to be AI literate, one needs to understand how to use the internet safely.*

Lecturer7 *"AI literacy encompasses more than just coding; it also includes ethical awareness, data fluency, tool proficiency, conceptual understanding of AI and machine learning, and the capacity to convey AI concepts across disciplines. It equips graduates to use and assess AI systems critically"*.

The ICT Managers pointed out that AI literacy includes the ability to analyse not only the AI systems but also the societal impacts. They also underlined the ethical responsibilities of the future workforce. One ICT Manager mentions clear and concrete skills. At the same time, he points out a significant skills gap among graduates, who might know the theory of AI but cannot apply it in practice, for example, in model optimisation and data security management. Participants associated this gap with limited opportunities to apply theoretical knowledge in practical contexts. It was asserted:

ICT Manager1 *"In the age of AI, literacy entails the capacity to evaluate AI systems and their societal implications critically."*

ICT Manager6 "*AI literacy means understanding how to work within computational constraints, knowing when to use cloud services vs. local processing, optimising AI models for limited hardware, understanding data management and storage implications, recognising security and privacy concerns with AI systems*"

The student's viewpoint on AI literacy offers a more detailed perspective, where participants revealed that technical components should be integrated with ethical and practical considerations. Students not only pointed out the need to understand AI's limitations and capabilities but also the importance of using pre-trained models to develop a continuous-learning attitude that would help them keep up with the rapid pace of AI change. The students consider AI literacy to be the combination of conceptual

knowledge, critical evaluation, and responsible use of existing AI tools. Thus, they strongly advocated for the use of practical machine learning libraries, real-world datasets, data preprocessing, ethical considerations, and model deployment.

Student3 *"I consider myself digitally literate; nonetheless, AI represents a higher echelon. We utilise ChatGPT and Gemini, but do we truly comprehend them? What are the risks? I believe 'AI literacy' should encompass more than merely understanding how to input a prompt."*

Student4 *"AI literacy means understanding what AI can and cannot do, even if I cannot implement everything myself. It's about: Knowing when a problem might benefit from AI approaches, understanding the basics of how machine learning works, being able to use pre-trained models and APIs, recognising bias and ethical issues in AI systems, and being able to learn new AI tools as they become available."*

Student5 *"Many students initially viewed AI as an effortless solution for coding tasks, only to discover a significant gap in their understanding of the underlying code, prompting a re-evaluation of their learning approach. It is making us reset and focus on truly grasping the fundamentals, not just getting the answer."*

5.2. Theme 2: gaps in curricula: current CS curriculum vs. industry needs

Data analysis revealed gaps and areas for improvement in the current CS curriculum relative to industry needs. These were the strengths and weaknesses of the current curriculum.

5.2.1. Strengths of the current curriculum

The current CS curriculum is widely recognised for its strong foundational components, which remain relevant to the evolving AI landscape and are broadly aligned with industry expectations. Across students, lecturers, and ICT managers, particular emphasis is placed on the grounding in core programming languages such as Java, C++, and Python, which supports the development of problem-solving capabilities. Foundational subjects, including discrete mathematics, data structures, and algorithms, were consistently described as providing the analytical foundation necessary to understand computational logic and AI-related concepts. Participants also noted that working within resource-constrained environments often strengthens students' adaptability, troubleshooting ability, and collaborative skills.

Student4 *"We get solid basics in programming (Java, C++, Python). The Mathematics foundation is quite strong; we do a lot of calculus and discrete math. Problem-solving skills are emphasised because we often have to work with limited resources."*

Lecturer6 "The *current foundational knowledge, such as Discrete Mathematics, Data Structures and Algorithms, and Principles of Programming Languages, provides essential building blocks for understanding AI models and computational logic. It also exposes students to AI concepts courses such as Artificial Intelligence, Expert Systems, Decision Support Systems, and Pattern Recognition and Image Processing, introducing students to core AI techniques and applications."*

ICT Manager4 *"The current Computer Science curriculum gives students a good developmental foundation to tap into AI fundamentals and build on this for future developments. The curriculum has introduced a program in informatics that allows one to build a strong foundation in data science"*.

ICT Manager6: *"Students develop strong troubleshooting skills because they constantly work with limited resources. They learn to optimise code and manage memory efficiently due to hardware constraints. Strong collaborative skills develop naturally due to resource sharing."*

These perspectives collectively indicate a shared view that the existing curriculum provides a solid theoretical and problem-solving foundation for further development of AI-related competencies.

5.2.2. Weaknesses of the current curriculum

Participants identified several limitations in the current Computer Science (CS) curriculum, particularly relating to limited practical exposure, insufficient AI-focused content, and a disconnect between theoretical instruction and applied competencies. Across stakeholder groups, concerns were raised about the

curriculum's strong emphasis on foundational theory with limited opportunities for students to engage in hands-on AI development, experimentation, and real-world problem-solving.

Lecturers and ICT managers perceived gaps in the integration of contemporary AI topics, tools, and frameworks, noting that existing modules often do not reflect current industry practices. The curriculum was described as lacking structured exposure to areas such as machine learning workflows, data preprocessing, design patterns, model deployment, and ethical considerations in AI.

ICT Manager6: "*Our servers cannot handle contemporary AI workloads-8GB RAM machines are being used in these data science classes. We continue to use outdated development tools because their licenses are expensive. Students are unable to access any cloud-based AI platform due to bandwidth limitations. I also see students graduating without ever training a machine learning model.*"

ICT Manager5 "*Students don't understand computational requirements of different AI models, no experience with distributed computing or parallel processing, limited understanding of database optimisation for large datasets, poor grasp of network implications for AI applications and a lack of understanding and application of design patterns*"

Lecturer7 "*It lacks comprehensive AI content, up-to-date tools, practical exposure, and real-life integration into the different disciplines that can be affected by AI. There is also little or no infrastructure to support AI learning, such as access to GPUs and large datasets*".

Students similarly described a gap between the largely theoretical curriculum and the practical AI competencies expected in industry. This gap was frequently linked to limited exposure to contemporary AI tools and frameworks, as well as insufficient opportunities to work with real-world datasets and applications. As a result, some students perceived themselves as insufficiently prepared for industry requirements upon graduation.

Student1 "*Internship prerequisites often need proficiency in specific frameworks, cloud platforms, and comprehension of business challenges. It appears we are perpetually in a state of catching up. Even lecturers who take up AI courses are insufficiently upskilled. It is all theory.*"

Student4 "*AI/ML is barely covered - maybe one elective course if lucky. Our lecturers often have not worked with modern AI tools themselves. We have no access to cloud computing resources or expensive software licenses. Internet connectivity issues make it hard to access online AI courses or datasets. Equipment is old, and those that they claim are new struggle with basic data processing*"

Student2 "*To maintain competitiveness, we require basic statistics and probability, which are currently being taught theoretically; Hands-on experience with machine learning libraries like scikit-learn; understanding of data preprocessing and cleaning; Experience working with real datasets, not just toy examples; Knowledge of AI ethics and bias issues; Ability to deploy models, even simple ones*"

These accounts collectively reveal a perceived imbalance between strong theoretical grounding and limited opportunities for applied, practice-oriented AI learning within the current curriculum.

5.3. Theme 3: barriers to AI-oriented curriculum reform

Participants consistently identified obstacles hindering the seamless integration of AI knowledge into the CS curriculum.

5.3.1. Lecturer preparedness and resistance to change

Participants across all three stakeholder groups described lecturer preparedness as a constraint in the transition towards AI-oriented curriculum reform. Lecturers noted uneven staff readiness and limited exposure to recent AI subfields, while ICT managers emphasised the reskilling demands associated with shifting from conventional digital literacy to AI-focused teaching. Some participants noted that the transition may be challenging for staff who have long relied on established teaching approaches, particularly where significant upskilling and adaptation are required. Students similarly perceived that some lecturers

had not yet updated their knowledge in line with current industry developments. These views are illustrated in the following extracts:

Lecturer6 *"We have ill-equipped lecturers who resist change, which will be difficult for curriculum design and integration. They have taught the same course for 10 years using the same notes, hence personalising them."*

ICT Manager4 *"Change is always resisted largely because there will be a need for reskilling and retooling. The soft side of things also presents a challenge, and long-established lecturers may find the transition to AI-oriented teaching difficult without substantial reskilling. This all contributes to factors that promote resistance to change."*

Student3 *"Lecturers believe that a computer science degree guarantees ample work prospects, so they do not want to upskill. This is increasingly inaccurate, as the labour market grows, it is significantly more competitive for graduates. This uncertainty introduces an additional challenge for us as students striving to prepare adequately, yet the lecturers have not fully upskilled."*

These accounts indicate that participants viewed lecturer preparedness and perceptions of change as key factors influencing the pace and effectiveness of AI-related curriculum reform.

5.3.2. Institutional bureaucracy and slow curriculum cycles

Participants identified institutional processes and the pace of curriculum revision as constraints affecting the timely integration of AI-related content. Lecturers described the curriculum change process as one that must pass through multiple layers of institutional approval, which can result in delays. They also noted that changes to institutional culture may be more challenging than adjustments to curriculum content. ICT managers similarly referred to the slow pace of curriculum updates, expressing concern that by the time revisions are implemented, industry requirements may have already evolved. Students also revealed the perceived gap between the pace of technological change and the rate of curricular updates.

Lecturer4 "*Each proposed modification must traverse multiple layers of bureaucracy, potentially hindering development. Changing institutional culture is often harder than changing the curriculum itself."*

ICT Manager3 "*Universities have a sluggish pace of progress. We will inform them of the necessity for graduates to have X, Y, and Z competencies; by the time the curriculum is revised, we will have progressed to A, B, and C. The agility is insufficient to keep pace with technological advancements."*

Student2 "*Our curriculum resembles a massive vessel that requires considerable time to change direction. Although new AI developments are frequently discussed, the courses do not evolve at the same pace. I heard about the curriculum change in my second year, went for an attachment year (the third year), and am now completing my fourth year. Still, the curriculum hasn't changed.*"

These perspectives indicate a shared perception that institutional processes and curriculum revision cycles may not always align with the rapid pace of change in AI-related technologies.

5.3.3. Resource constraints and unequal access

Participants identified resource constraints and unequal access as underlying systemic barriers affecting the adoption and effective implementation of AI within the curriculum. While the previous theme highlighted limitations in curriculum design and delivery, participants in this context emphasised that broader infrastructural and access-related conditions constrain both the feasibility of teaching and student participation in AI-related learning.

Lecturers described the challenge of teaching AI without adequate training, tools, and institutional support, while ICT managers pointed to hardware limitations, software costs, and infrastructural instability as ongoing constraints. These included limited access to high-performance computing resources such as GPUs, restricted availability of cloud platforms, high licensing costs for AI tools, unreliable internet connectivity, and power instability, all of which were seen as limiting the feasibility of delivering practical AI instruction.

Lecturer3 *"We want to teach AI, but without proper training and resources, it's daunting."*
ICT manager6 *"Our best machine has 16GB RAM, insufficient for most modern AI training; Commercial AI tools are beyond our budget; Unreliable internet makes cloud-based learning frustrating; Frequent outages interrupt long-running AI training processes; Large datasets are difficult to store and share."*

Students also highlighted disparities in access to devices and internet connectivity, underscoring a persistent digital divide that impedes their ability to engage effectively with AI-related coursework. Limited access to personal computers, reliance on basic smartphones, and inconsistent internet access were described as barriers to participation, particularly in resource-intensive AI learning environments.

Student4 *"There is a serious digital divide: Students with no home internet access fall behind in AI coursework. There are also device limitations: Many students only have basic smartphones or old laptops. We also need technical support: Students need more help with AI tools than traditional programming."*

These accounts indicate that resource availability and access conditions shape both the implementation of AI-focused instruction and the extent to which students can meaningfully participate in AI-related learning activities.

5.3.4. Curriculum overload and preservation of foundational knowledge

Participants described a tension between integrating AI-related content and maintaining existing foundational components within the CS curriculum. Lecturers highlighted the limited capacity of an already full programme, noting that the inclusion of additional topics such as AI, ethics, and data literacy must be balanced against the need to preserve core programming and computational skills. Some participants also expressed concern that increased reliance on AI tools could undermine the development of fundamental coding skills. Students similarly described the curriculum as already comprehensive, raising questions about how new content could be incorporated without increasing the workload or programme duration.

Lecturer2 *"We are attempting to incorporate AI; however, it is challenging to identify a suitable space. The curriculum is already full of these MBKs(Minimum Body of Knowledge). Students frequently encounter the challenge of overdependence on these tools, raising fears that their fundamental coding abilities may deteriorate."*
Student5 *"Incorporating new content necessitates the removal of existing material. Our curriculum is already comprehensive. How can they further incorporate AI, ethics, and data literacy into the curriculum without increasing the difficulty or duration of the degree program?"*

These accounts indicate a shared concern about balancing the introduction of AI-related competencies with the preservation of essential foundational knowledge within existing curriculum structures.

5.3.5. Limited African-centred AI content

Participants highlighted the limited presence of African-centred AI content as a challenge in the current curriculum. Across stakeholder groups, concerns were raised about the reliance on datasets, case studies, and ethical frameworks developed predominantly in Western contexts, which may not fully reflect local realities. Lecturers noted that this may limit students' opportunities to engage with contextually relevant problems, while students expressed a need for AI applications that address local challenges and incorporate African languages, data, and socio-economic contexts.

Lecturer2 *"Most AI resources, case studies, AI training data and curricula are derived from Western contexts, so our students are learning to solve problems that may not be pertinent to their communities or with data that doesn't account for the subtleties of African language and culture. This severely restricts their capacity for local innovation, leaving a void in relevant applications for African students."*
Student3 *"European or Silicon Valley examples are nearly always used when we learn about AI. We must examine how AI can be applied to our languages and data to address issues in local governance,*

healthcare, and agriculture across Africa. Otherwise, it seems we are merely importing solutions rather than coming up with our own".

These accounts indicate a shared perception that greater inclusion of locally relevant data, contexts, and applications may enhance the relevance of AI education within African higher education settings.

5.4. Theme 4: strategies for reimagining the curriculum

Thematic analysis revealed that a future-proof curriculum must be collaborative, interdisciplinary and project-based.

5.4.1. Stronger industry-university collaboration

Participants from all stakeholder groups emphasised the importance of collaboration between academia and industry to reimagine the CS curriculum. This collaboration was described as enabling access to practical insights, real-world datasets, and industry-relevant experiences that may support the integration of AI-related competencies. Participants also highlighted that such partnerships could help align curriculum content with evolving labour market expectations.

ICT managers described collaboration as involving clearly defined roles across stakeholders. One participant outlined a "7 Pillars for a successful AI adoption process," in which universities focus on skills development and talent preparation, while industry contributes by providing data, computing resources, and exposure to market needs. The role of government was also noted as supporting governance and coordination within this collaborative structure.

ICT Manager6 *"We can offer internships, provide insights on emerging industry trends, co-design capstone projects, and even contribute to curriculum design discussions."*

Lecturer5 *"We need industry leaders to come in and give guest lectures, share real-world datasets, and even mentor students on capstone projects. It bridges the gap between what we teach and what is needed, making the benefits of the redesigned curriculum more observable."*

Lecturer6 *"Industry's role should encompass the provision of real-world context and data, facilitating experiential learning, provision of resources and funding, whilst for universities they should focus on Curriculum design and Pedagogy, research and futureproofing, as well as faculty development and training"*

Student2 *"The industry should work with universities by giving feedback, offering internships, and helping students learn the latest tools and needs. Both should work together to prepare students for jobs".*

These accounts indicate a shared view that structured collaboration among universities, industry, and other stakeholders may help align computer science curricula with emerging AI skill requirements.

5.4.2. Interdisciplinary curriculum restructuring

Participants described the need to move beyond disciplinary silos and incorporate AI-related concepts across multiple fields. This was framed as requiring an interdisciplinary approach, in which AI is understood not only within traditional computer science domains but also in relation to ethics, law, the humanities, and the social sciences. Lecturers and ICT managers highlighted the importance of restructuring curricula to support cross-disciplinary learning through both the introduction of new modules and the adaptation of existing courses.

Lecturer8 *"Integrating interdisciplinary themes into core courses can enhance student engagement and foster innovative thinking. The curriculum can be made more relevant and engaging for students by integrating fields such as data science, psychology, and environmental studies.*

Lecturer4 *"The reimagined curriculum needs to be less siloed. We need more interdisciplinary projects, particularly in ethics, the social sciences, and law. AI is not just code; it has a societal impact."*

ICT Manager5 *"We need to break down silos between departments and encourage collaborative projects that span multiple disciplines to prepare students for the realities of the job market."*

These accounts indicate a shared perspective that interdisciplinary integration may support a more holistic understanding of AI and its applications across different domains.

5.4.3. Project-based and problem-based AI learning

Participants consistently highlighted project- and problem-based learning as important approaches for embedding AI within the computer science curriculum. This approach was described as supporting the connection between theoretical knowledge and practical application, enabling students to engage with real-world AI problems and complex ("messy") datasets. Across stakeholder groups, there was an emphasis on developing applied AI literacy, in which students move beyond conceptual understanding to build, analyse, and evaluate AI-driven solutions. Such forms of engagement were also associated with opportunities to collaborate with industry, providing exposure to real-world contexts that may not be fully replicated in traditional classroom settings.

Student1 *"More practical labs and real-world case studies would be great. Perhaps even opportunities to collaborate with real companies on AI challenges. That helps us connect the theory to what we'll be doing and makes the learning feel more tangible and useful for our careers."*

Lecturer4 *"Engaging students through real-world projects allows them to apply their knowledge and see the impact of their work firsthand."*

ICT Manager1 *"Project-based learning with real-world AI problems is key. Give students messy data and ask them to identify a business problem that AI can solve. Then, have them build and critically evaluate the solution. That's how they learn to apply AI literacy, not just define it."*

ICT Manager3*" Partnerships can provide students with invaluable insights and experiences that a traditional classroom setting cannot."*

These accounts indicate a shared view that experiential and problem-based approaches may support the development of applied AI competencies within the curriculum.

5.4.4. Ethical, applied, and context-responsive AI competencies

Participants, particularly ICT managers, described a shift in the competencies expected of computer science graduates in the context of AI. Emphasis was placed on applied problem-solving, ethical reasoning, and the ability to evaluate AI outputs, alongside traditional programming skills. Participants also highlighted the importance of understanding AI systems' limitations and of communicating AI-related concepts to both technical and non-technical stakeholders. In addition, competencies related to data literacy, cloud management, and the practical application of AI in business and local contexts were identified as increasingly relevant.

ICT Manager1 *"An AI-literate graduate understands the limitations of AI, its ethical boundaries, and how to communicate complex AI concepts to non-technical stakeholders. They also need strong data literacy, specifically for AI models. This competence is crucial for their perceived usefulness in our teams."*

ICT Manager2 *"Industry leaders anticipate that AI assistants will soon achieve a proficiency level comparable to that of mid-level software engineers. This fundamentally reshapes the essential skills we seek; it's less about rote coding and more about problem-solving, ethical reasoning, and critical evaluation of AI outputs."*

ICT Manager4 *"AI has developers of AI tools and users of AI tools. The Computer Science graduate should focus on developing AI tools, which can be used as business solutions. To achieve these key areas, AI tools programming, Data Lake platform development, Cloud management, AI governance, and ethics should be considered"*

These accounts indicate a shared perspective that AI literacy extends beyond technical proficiency to encompass ethical awareness, contextual understanding, and the ability to apply AI in practical, domain-specific settings.

5.4.5. Resource-aware implementation models

While the previous theme (resource constraints and unequal access) highlighted limitations in curriculum design, this theme focuses specifically on systemic and infrastructural constraints that affect implementation. Participants described the need for resource-aware approaches to support AI implementation within the curriculum, particularly in contexts with infrastructural constraints. ICT managers emphasised flexible models that extend beyond traditional lab-based setups, incorporating a combination of technological and support mechanisms. A hybrid computing approach was frequently highlighted, combining local processing for foundational tasks with cloud-based resources for more advanced AI workloads. Participants also noted the relevance of a mobile-first approach, recognising that many students primarily access learning through smartphones, alongside the use of offline-capable materials and low-bandwidth tools.

ICT Manager6 *"Infrastructure requirements include: Hybrid computing model: Combination of local processing for basics and cloud for advanced work; Mobile-first approach: Many students only have smartphones, so curriculum should accommodate this; Offline-capable resources: Materials that work without constant internet connectivity; Energy-efficient computing: Solar-powered labs and low-power computing solutions"*

Participants further emphasised the importance of using open-source software, free cloud services, and energy-efficient solutions to accommodate existing infrastructure limitations. The role of ICT managers was also highlighted in supporting implementation through technical support, infrastructure planning, and optimising available resources. In addition, considerations around data security and responsible management of AI tools were noted as part of the broader implementation process.

ICT Manager1 *"Universities should invest in reliable internet connectivity. It's more important than expensive hardware. Focus on open-source tools and free cloud resources, build partnerships with international institutions for resource sharing, train a core team of students and faculty who can support AI initiatives"*

These accounts indicate a shared view that effective integration of AI into the curriculum depends not only on curriculum content but also on the availability of adaptable, accessible, and well-managed technological environments.

5.4.6. Successful AI literacy initiatives

Participants highlighted examples of existing initiatives that support the development of AI literacy, particularly within African contexts. Across stakeholder groups, emphasis was placed on the role of policy frameworks, accessible learning resources, and community-driven approaches in supporting AI education. ICT managers referred to continental and international frameworks, such as UNESCO guidelines and the African Union Digital Education Strategy, as providing direction for the responsible and structured integration of AI into education systems. National-level initiatives, such as Rwanda's AI policy, were also noted as examples of efforts to guide both development and risk mitigation.

ICT Manager4 *" … UNESCO released competency frameworks for students and lecturers to guide them on AI use and misuse in the classroom. The African Union Digital Education Strategy and Implementation Plan has established a framework for engagement and the acceleration of digital technologies to align with Africa's digital education. … … Rwanda is the first African country to have an AI policy that acts as a roadmap to harness the benefits of AI and mitigate its associated risks, especially those associated with sharing children's data."*

Participants also described practical, community-based initiatives that provide accessible pathways to AI learning. These included boot camps, hackathons, online training platforms, and collaborative programmes that emphasise hands-on learning and real-world problem solving. Several examples were provided of initiatives that utilise open-source tools, free cloud resources, and offline or low-bandwidth solutions to accommodate infrastructural limitations.

ICT Manager6 *"Google for Education credits: Free access to Google Cloud Platform for student projects; Jupyter notebooks on local servers: Providing web-based AI development environment; Offline AI resources: Downloaded datasets and pre-trained models for when the internet fails; Mobile learning apps: Simple AI concepts taught through smartphone apps; Community partnerships: Local businesses providing real datasets for student projects*

Student5 *"Some universities now offer AI bootcamps or hackathons where students learn by doing real projects. Google's AI training materials are also helpful and easy to follow. Wethink_code South Africa also offers courses related to that.*

Lecturer7 *"Initiatives like Zindi.Africa, Data Science Nigeria, AI Saturdays, DeepLearning.AI, and AI4D demonstrate scalable, community-driven approaches to AI education in Africa. These models emphasise real-world problem solving, peer learning, and accessibility, offering workable templates for Zimbabwe".*

These accounts indicate a shared view that a combination of policy support, accessible learning pathways, and collaborative initiatives may help advance AI literacy in resource-constrained contexts.

6. Discussion

This qualitative study aimed to navigate the critical shift from digital to AI literacy and its implications for reimagining the CS curriculum, drawing insights from lecturers, ICT managers, and students.

6.1. From digital literacy to AI literacy

The Constructivist Learning Theory establishes the gap between the intended pedagogy and the orientation of students' learning. The interviews revealed that most students view AI literacy as a set of discrete technical skills, requiring memorisation and an objectivist stance, shaped by years of formal, rote instruction. This, however, stands counter to the constructivist principles underlying the proposed project-based learning model (Vorsah & Oppong, 2024). Project-based learning and real-world case studies align with contemporary pedagogical traditions that support active and applied learning, thereby potentially enhancing the trialability and observability of AI concepts in the curriculum (Voulgari et al., 2021). Thus (Fakhar et al., 2024), proposes that project-based and vertically integrated learning should gradually scaffold ethical and technical AI across all study years, culminating in complex final-year projects in which the basic "why" is integrated with the "how" of applied AI. Equally important are periodic curriculum reviews and continuous faculty development programs, which enable lecturers to effectively impart a blend of AI and CS content (Fakhar et al., 2024; Sun et al., 2023). Together, these strategies provide a sustainable, future-ready curriculum that maintains academic rigour while ensuring relevance in an AI-driven educational landscape. Additionally, the literature proposes Game-Based and Data-Driven Learning, as well as the use of personally meaningful data, to engage students and clarify abstract AI concepts (Voulgari et al., 2021). The curriculum redesign, hence, must address underlying epistemological orientations by promoting collaborative, problem-centred approaches that shift students from passive knowledge acquisition towards active construction of AI knowledge. This aligns with the results, which consistently identified project- and problem-based learning as mechanisms for bridging the gap between theoretical instruction and practical AI application, particularly when working with real-world, "messy" datasets. Increasingly, the emphasis is shifting from preparing graduates to compete against AI systems towards equipping them to collaborate effectively with AI technologies in human-AI partnership environments. This emerging orientation positions AI literacy not merely as technical proficiency, but as the capacity to critically supervise, evaluate, and strategically augment AI-generated outputs within complex socio-technical contexts.

While the results captured perceptions of positive AI literacy, a more critical examination revealed tension lurking beneath. For example, through the TAM and DOI framework, while students and industry professionals considered AI tools and competencies useful, lecturers expressed reservations about their immediate compatibility with learning contexts, indicating a divergence between their theoretical acceptance of AI and its real-world adoption. Such a disagreement aligns well with the TAM, requiring not only consideration of the perceived usefulness of the innovation but also practical acceptance

barriers (Ajibade, 2018). The observed disparity between high PU and low PEOU indicates that, in the absence of systematic professional development and institutional support, mere recognition of AI's pedagogical value will not translate into meaningful adoption (Fakhar et al., 2024; Ajibade, 2018). This is further reflected in the results, where lecturer preparedness, limited exposure to contemporary AI tools, and perceived resistance to change were identified as key barriers to curriculum reform. The traditional TAM assumption of rational uptake following perceived usefulness is therefore challenged in this context.

A recurring concern across the findings was the mismatch between the present CS curricula and the skill set demanded by the industry. Participants consistently indicated that graduates lack practical experience and applied technical competencies, a conclusion that resonates with the one drawn by (Iniesto et al., 2021). This reinforces broader evidence of skill gaps within African ICT sectors, where traditional instructional approaches lag behind industry demands, particularly in areas such as Python-based ML frameworks (Chumwatana & Hpone, 2025). The results further highlighted that students often perceive themselves as "catching up" with industry expectations due to limited exposure to real-world tools, cloud platforms, and deployment environments. The urgency to bridge the gap is evident; lecturers must align curricular experiences with hands-on applications of AI in real-world scenarios. Ensuring alignment between educational offerings and the practical skills employers demand is paramount to students' employability (Faustino et al., 2025). Consequently, curriculum transformation must extend beyond the assumptions of DOI theory, requiring deliberate alignment with labour market demands and evolving technological ecosystems.

The DOI perspective enabled a nuanced analysis of adoption, accounting for lecturers' reflections on issues of resistance, institutional inertia, and uneven implementation. While relative advantage (employability) and observability (success of AI-integrated programs elsewhere) were rated high by the participants, trialability remained low, and complexity was high (Rogers et al., 2019). The high complexity arises from issues such as limited faculty upgrading, a shortage of African-centric cases, and infrastructural problems (Journal & Maluleke, 2025). As reflected in the results, institutional bureaucracy and slow curriculum revision cycles further constrain responsiveness, with participants describing curricula as evolving more slowly than the pace of technological change. Addressing these barriers requires strategic planning and institutional support to execute these changes, which may be warranted given their complexity. These problems mirror those reported in the literature and expose systemic issues in educational environments, thereby blocking the adoption of new technologies (Ibrahim et al., 2022). To maintain relevance and avoid lagging on the adoption curve, an institution should be open to such advancements. In practice, contemporary resources should align with industry needs.

The findings also demonstrate that integrating AI into existing curricula cannot be achieved solely through additive approaches, as this can lead to curriculum overload. Achieving balance requires a conscious effort to use contextual integration, along with modularisation and strategic pruning, aligning with the suggestions by (Torró, 2025; Chen et al., 2023). Foundational subjects such as Algorithms and Data Structures can be recontextualised through AI applications, for example, by analysing algorithmic efficiency using neural network training data or evaluating computational resource efficiency in data-intensive environments (Khan et al., 2025). Theories taught this way thus transition from abstract theoretical instruction to applied, practical learning. In parallel, universities must strategically carve out curricular space by eliminating outdated material and methods, such as an intensive focus on legacy programming syntax, and by introducing more relevant modules taught at a high level, such as data wrangling with Python, cloud computing, and model design patterns and deployment. Modularisation with electives can enable students to continue solid foundations in CS and, at the same time, take specialised AI-focused pathways (Ma et al., 2025). Interdisciplinary integration could also lessen cognitive burden while enhancing students' holistic appreciation of AI's societal impact, including AI ethics in social science/business. This resonates with DoI Theory principles, particularly those regarding the compatibility of new ideas with existing structural conditions (Rogers et al., 2019). The literature, moreover, commonly cites the need to embed AI literacy across disciplines and while integrating technical, ethical, and societal dimensions for meaningful learning (Baskara, 2025; Casal-Otero et al., 2023). This is further reinforced by findings on the need for interdisciplinary curriculum restructuring and the inclusion of contextually relevant AI applications within African higher education.

6.2. Why curricular reform remains difficult

Beyond formal curriculum reform, achieving truly all-encompassing and current AI literacy requires recognising and integrating non-curricular learning pathways (Pendergast et al., 2024). Given the rapid pace of change in AI technologies, universities can no longer serve as the sole source of knowledge but must increasingly act as facilitators of lifelong learning (Preece, 2017). This calls for a system in which professional certifications, such as those offered by Google, IBM, and Microsoft, are mutually recognised, credited, and considered legitimate forms of learning that provide industry-relevant, continuously updated skills (Veeneman, 2018; Kovalev et al., 2025).

Moreover, access to curated open educational resources, including technical videos, podcasts, and tutorials, can support the democratisation of AI knowledge. Equally important are opportunities for hands-on experience, such as hackathons, innovation challenges, and virtual lab environments, such as Kaggle or Google Colab, where students interact with real datasets and apply problem-solving scenarios (Sajja et al., 2024; Beldar et al., 2025). Mentorship programmes that connect students with AI professionals and alumni further strengthen these learning pathways (Chi et al., 2012). By embodying these different ways of learning, the universities, in turn, transition from being repositories of knowledge to facilitators of continuous, practice-oriented learning.

However, the findings of this study indicate that the difficulty of curricular reform extends beyond the availability of alternative learning pathways. Structural and institutional constraints identified in the results, such as limited lecturer preparedness, resistance to change, and slow curriculum revision cycles, continue to restrict the pace at which formal curricula can adapt to emerging AI competencies. Participants described curriculum reform processes as lengthy and bureaucratic, often lagging the rapid evolution of AI technologies.

In addition, resource constraints and unequal access further complicate reform efforts. While alternative learning pathways such as online platforms and cloud-based tools offer flexibility, their effective use remains uneven due to infrastructural limitations, including unreliable internet connectivity, limited access to computing resources, and disparities in students' access to devices. These conditions create uneven opportunities for engagement with AI learning, reinforcing the gap between theoretical exposure and practical competence.

The findings also highlight tensions related to curriculum overload and the preservation of foundational knowledge. Integrating AI into existing programmes is not simply a matter of adding new content, as this risks increasing cognitive load and diluting core computer science principles. As a result, reform requires careful restructuring rather than incremental expansion, further complicating the process.

Collectively, these factors suggest that curricular reform in the AI era is not solely a pedagogical challenge, but a multi-layered institutional process shaped by organisational, technological, and socio-economic constraints. This helps explain why, despite widespread recognition of the importance of AI literacy, its systematic integration into computer science curricula remains uneven and gradual.

While these barriers are presented across distinct themes in the results, they should not be interpreted as isolated challenges. Rather, they reflect interconnected institutional constraints that collectively shape the pace and feasibility of AI curriculum reform

6.3. What a resource-aware reimagined curriculum requires: strengthening regional context and evidence base

Observations identified in this study were interpreted against the existing bodies of literature in triangulation. Such literature includes recent African works that interrogate and theorise AI literacy and curriculum change in higher education, as well as regional initiatives such as the African Union Continental Strategy for Education (CESA 2016–2025) (African Union, 2015; Awaah, 2024) and Zimbabwe National AI Strategy (2026–2030) (MICTPCS\) Ministry of ICT, 2026). Many of the issues emerging from these sources are congruent with the themes raised by the participants, in particular institutional readiness gaps, uneven faculty digital capacity, and the urgent need for ethical and localised AI integration. For instance, participants' notes about the absence of AI-integrated modules were corroborated by an analysis of the course outline, which revealed limited or no structured inclusion of AI literacy components. Similarly,

policy documents highlight persistent challenges, including insufficient funding and limited staff capacity, reinforcing the structural nature of the barriers identified in this study.

Beyond validation, the findings extend existing regional evidence by illustrating how these structural constraints manifest in practice. Participants described a learning environment characterised by infrastructural limitations, limited access to cloud-based platforms, and unequal student access to digital resources, collectively shaping the feasibility of implementing AI curricula. These insights reinforce the need for resource-aware curriculum models that are not only pedagogically sound but also contextually adaptable.

In addition, the findings emphasise the importance of embedding African-centred AI content within curricula. As participants highlighted, reliance on Western datasets, case studies, and ethical frameworks limits the relevance of AI education to local contexts and constrains students' ability to develop solutions to region-specific challenges. This aligns with broader concerns in the literature about the risk of reproducing digital dependency when local knowledge systems and socio-economic realities are not adequately represented.

The study further demonstrates that effective curriculum reimagining requires a shift from purely content-driven reforms towards integrated approaches that combine curriculum restructuring, pedagogical innovation, and institutional support mechanisms. This includes interdisciplinary integration, project-based learning, and stronger industry–university collaboration, all of which were identified in the results as critical enablers of applied AI literacy.

Taken together, these findings suggest that a reimagined computer science curriculum in resource-constrained contexts must be both adaptive and context-sensitive. Rather than replicating models from well-resourced environments, institutions must adopt flexible, resource-aware strategies that leverage hybrid infrastructures, open-source tools, and alternative learning pathways. This strengthens the regional evidence base by providing empirically grounded insights into how AI literacy can be meaningfully integrated within African higher education systems. These resource-aware approaches also contribute to the emerging discourse on sustainable AI education, particularly within resource-constrained contexts where equitable access, energy efficiency, and long-term infrastructural viability remain critical considerations.

6.4. Localising AI literacy in African higher education: ethical and contextual dimensions

Participants' observations on strengthening the ethical dimension suggest that a growing regional scholarship disadvantages the applied Global North-trained AI model. A significant emphasis is placed on the necessity of teaching data sovereignty to counter algorithmic bias in African institutions, where gaps exist in local cultural and linguistic data (Bosarge, 2024). This issue is not one of preference; it is a systemic failure to ensure contextual relevance, thereby blocking the application of an AI literacy framework suited to African higher education. When case studies, datasets, and pedagogical models are predominantly of Global North origin, curricula unconsciously guide graduates to become customers of AI solutions rather than creators grounded in their own communities. Such epistemic dependency is termed "algorithmic colonialism" (Patra, 2025), wherein African realities are marginalised in knowledge production. The ethical considerations are significant: the non-availability of locally representative datasets restricts the region's ability to detect and mitigate algorithmic biases related to language, demographics, and socio-economic context, thereby reproducing digital colonialism and undermining the emancipatory promise of AI (Mhlanga, 2023; Van Wyk & Attribution, 2024). These concerns were strongly reflected in the Results, where participants highlighted the limited relevance of Western-centric AI examples and the need for locally grounded applications in areas such as governance, healthcare, and agriculture.

The absence of locally made content significantly increases the complexity that lecturers consider when reviewing DOIs, as they are required to spend substantial, often unpaid, time preparing locally relevant examples and datasets, which creates a significant internal barrier to adoption (Journal and Maluleke, 2025). Under the TAM model, the absence of African context materials further reduces the perceived use of AI-oriented curricula, as both faculty and students find it difficult to relate imported content to indigenous societal needs. CLT provides another perspective: when learners cannot locate AI principles within their own linguistic or cultural frames of reference, the possibilities for social knowledge

construction and meaningful engagement are significantly reduced. Thus (Lewis, 2025), argue that in emerging-technology education, knowledge construction is contingent upon authentic contexts, collaboration, and reflective practice. This aligns with findings on lecturer preparedness and resistance, where the burden of adapting content contributes to slower uptake and uneven implementation of AI-oriented curricula.

To address this imbalance, there is a significant dearth of African-centred AI content in the current computer science curriculum, necessitating efforts to decentralise AI education by incorporating ethical and contextual considerations, community-driven AI problem-solving scenarios, and locally relevant datasets. Breaking silos must thus enable stakeholders to view the curriculum beyond job preparation, toward equipping students to contribute meaningfully to society. The literature recommends that modular and adaptable curricula can be docked across disciplines rather than taught as independent subjects (Casal-Otero et al., 2023; Bosarge, 2024); this finding provides a counterpart scenario in the social realm of CLT, where shared understandings and solutions are socially constructed through interaction (Schwebel, 1979). Empirical findings from this study further emphasise the importance of interdisciplinary curriculum restructuring and project-based learning as mechanisms for embedding contextually relevant AI competencies. By forming partnerships among lecturers, industry professionals, and students, stakeholders can develop a more integrated and relevant curriculum to prepare students for diverse career paths (Casal-Otero et al., 2023; Tadimalla & Maher, 2024). This would, in essence, bring the abstract concept of technological data sovereignty into actual practice, while ensuring that AI literacy education aligns with the linguistic, ethical, and socio-economic fabric of African life. Enabling policy scaffolds for such localisation are provided by national and continental interventions, such as Zimbabwe's National Artificial Intelligence Strategy (2021) (MICTPCS\) Ministry of ICT, 2026) and the African Union's Data Policy Framework (2022) (African Union, 2022). Establishing these as resources upon which to build would transform African higher education from a passive adopter of imported technology into a proactive innovator of, and custodian for, a contextually relevant AI ecosystem. Participants in this study highlighted the need for stronger industry–university partnerships, resource-sharing mechanisms, and hybrid infrastructure models to support implementation in resource-constrained environments.

Ultimately, aligning empirical findings with theoretical interpretation and curricular action situates the absence of African-centric content not as a mere pedagogical oversight but as an ethical and epistemological challenge at the heart of the future of AI literacy in African higher education. This reinforces the need for a resource-aware, context-sensitive curriculum framework that integrates policy support, curriculum redesign, and institutional strategy.

Collectively, the barriers identified in this study should not be interpreted as isolated operational shortcomings but rather as interdependent constraints that influence the adoption of AI-oriented curricular reform. Lecturer preparedness, bureaucratic curriculum cycles, infrastructural limitations, and the lack of locally relevant AI content collectively increase the perceived complexity of reform. Interpreted through the DOI, these factors reduce trialability and slow diffusion; through the TAM, they lower perceived ease of implementation; and through the CLT, they weaken the collaborative and experiential conditions necessary for meaningful AI literacy development. This integrated perspective provides a foundation for the proposed framework, which seeks to address these constraints in a coordinated and contextually grounded manner.

7. Implications

The findings indicate that effective AI literacy reform requires alignment between institutional strategy, national policy, and evolving workforce demands. This qualitative study offers implications for multiple stakeholders within computer science education and beyond. Building on the proposed framework, this section outlines implications for curriculum design, institutional strategy, and policy development. These are grounded in the framework's three core dimensions: foundational-applied integration, ethical-contextual alignment, and pedagogical-collaborative transformation and its emphasis on staged, resource-aware implementation.

7.1. Implications for curriculum design

At the curriculum level, the findings highlight the need to move beyond the traditional separation between foundational computer science content and emerging AI competencies. The framework provides a structured pathway for embedding AI literacy within existing modules rather than treating it as an isolated component. This implies that curriculum designers should prioritise integration over replacement, ensuring that core concepts such as algorithms, programming, and data structures are preserved while enriched with applied AI use cases.

In practice, this requires reconfiguring course content to incorporate AI-related examples, datasets, and problem-solving tasks into existing modules, alongside introducing project-based and interdisciplinary learning experiences. The emphasis on ethical–contextual alignment further implies that AI education should include locally relevant case studies, data governance considerations, and critical reflection on bias and societal impact. This shifts curriculum design from a purely technical orientation towards a more holistic model of AI literacy that is contextually grounded and socially responsive.

7.2. Implications for institutional strategy

At the institutional level, transitioning towards AI-oriented curricula requires coordinated strategic action rather than isolated pedagogical adjustments. The framework emphasises staged adoption, suggesting that universities begin with targeted pilot initiatives such as AI-integrated modules or capstone projects and progressively scale these efforts based on institutional capacity and evaluation outcomes.

A key implication is the need for sustained investment in lecturer upskilling and capacity development. Institutions should prioritise continuous professional development programmes that equip academic staff with both technical AI competencies and pedagogical strategies aligned with constructivist and experiential learning approaches. In addition, institutional strategies must address infrastructural constraints through flexible, resource-aware solutions, including open-source tools, shared computing environments, hybrid cloud models, and mobile-accessible platforms.

The framework also highlights the importance of industry–academic collaboration. Universities should formalise partnerships with industry to co-develop curriculum content, provide access to real-world datasets, and support student learning through internships, mentorship, and project-based engagement. Such collaboration enhances curriculum relevance and facilitates smoother transitions into the labour market.

7.3. Implications for policy and governance

At the policy level, the findings point to the need for adaptive and forward-looking governance frameworks that support the rapid evolution of AI-related competencies. National and institutional policies should move beyond static curriculum approval processes towards more flexible mechanisms that allow timely updates to programme content. This further positions higher education institutions not only as sites of AI skills development, but also as critical actors in shaping responsible AI governance cultures within emerging digital economies.

The framework suggests that policymakers prioritise context-sensitive AI education strategies, particularly in resource-constrained environments. This includes supporting the development of locally relevant datasets, promoting open educational resources, and incentivising collaboration between universities, industry, and government. Policies should also address equity concerns by ensuring that under-resourced institutions are not excluded from AI-driven educational transformation.

Furthermore, regulatory frameworks should explicitly integrate ethical and data governance considerations into AI curriculum guidelines, aligning educational practice with broader societal priorities such as privacy, fairness, and responsible innovation. This ensures that graduates are both technically competent and ethically aware.

7.4. Ethical and societal implications in the African context

Embedding a strong ethical foundation in AI literacy is essential to ensuring that AI contributes to equitable, contextually relevant development in African higher education. The curriculum should extend

beyond general ethical principles to cultivate context-specific awareness, enabling students to critically examine how AI interacts with local realities.

Ethical training should address key issues such as algorithmic bias, data privacy, and the socio-economic implications of AI within African contexts. The reliance on datasets derived primarily from Western environments risks reinforcing linguistic and cultural exclusion, leading to models that are not well-suited to African languages, social structures, or decision-making contexts. Students should therefore be equipped to identify and mitigate such biases through inclusive data practices and locally grounded system design.

In addition, the curriculum should address concerns related to data colonialism by empowering students to safeguard local data resources, advocate for equitable benefit-sharing, and understand intellectual property rights. Integrating indigenous knowledge systems and local ethical frameworks into AI education ensures that technological development aligns with societal values and lived realities.

In contexts such as Zimbabwe, where socio-economic challenges remain prominent, ethical considerations should also extend to the design of AI systems that augment rather than replace human labour. This aligns with broader educational goals, such as Education 5.0, which emphasises human capital development and socially responsive innovation.

7.5. Proposed framework AI literacy model for reimagining the CS curriculum

The transition from digital literacy to AI literacy requires more than incremental curriculum updates; it calls for a structured reconfiguration of computer science education that is responsive to technological shifts, institutional realities, and socio-economic contexts. Drawing on empirical findings and interpreting them through the lenses of the TAM, DOI, and CLT, this study proposes a Resource-Aware AI Literacy Curriculum Framework (RA-AILCF) (Figure 1). The proposed framework integrates three dimensions: (1) foundational-applied integration, (2) ethical-contextual alignment and (3) pedagogical-collaborative transformation, implemented through a staged, resource-aware approach. Collectively, these dimensions are intended to support the development of graduates capable of engaging critically, ethically, and collaboratively within increasingly AI-mediated professional and learning environments

What distinguishes this framework from existing AI literacy and curriculum models is its explicit integration of pedagogical, contextual, and implementation dimensions within a single operational structure. Rather than treating AI literacy as a set of competencies or as standalone modules, the framework provides a phased, resource-aware pathway for curriculum transformation. It enables institutions to (i) embed AI concepts within existing foundational modules, (ii) progressively introduce applied and interdisciplinary learning components, and (iii) align implementation with institutional capacity and infrastructural constraints. This positions the framework as both a conceptual and practical guide for curriculum designers operating in resource-constrained environments.

1. **The Foundational-Applied Integration axis** emphasises that AI literacy should not displace core computer science principles such as algorithms, data structures, and systems design. Instead, AI concepts should be embedded within these foundational modules to ensure continuity while enhancing relevance. This approach reflects participants' concerns about maintaining disciplinary rigour while responding to evolving industry expectations.
2. **The Ethical-Contextual Alignment axis** foregrounds the importance of grounding AI education in local socio-cultural and regulatory contexts. This includes addressing issues such as data bias, privacy, and the relevance of AI applications within African settings. The empirical findings highlighted a clear need for curricula that move beyond imported content to incorporate locally meaningful datasets, case studies, and ethical considerations.
3. **The Pedagogical-Collaborative Transformation axis** reflects a shift from traditional, lecture-driven instruction towards more interactive, student-centred learning environments. In line with CLT, the framework advocates for project-based learning, interdisciplinary collaboration, and engagement with real-world problems, supported by stronger partnerships between academia and industry.

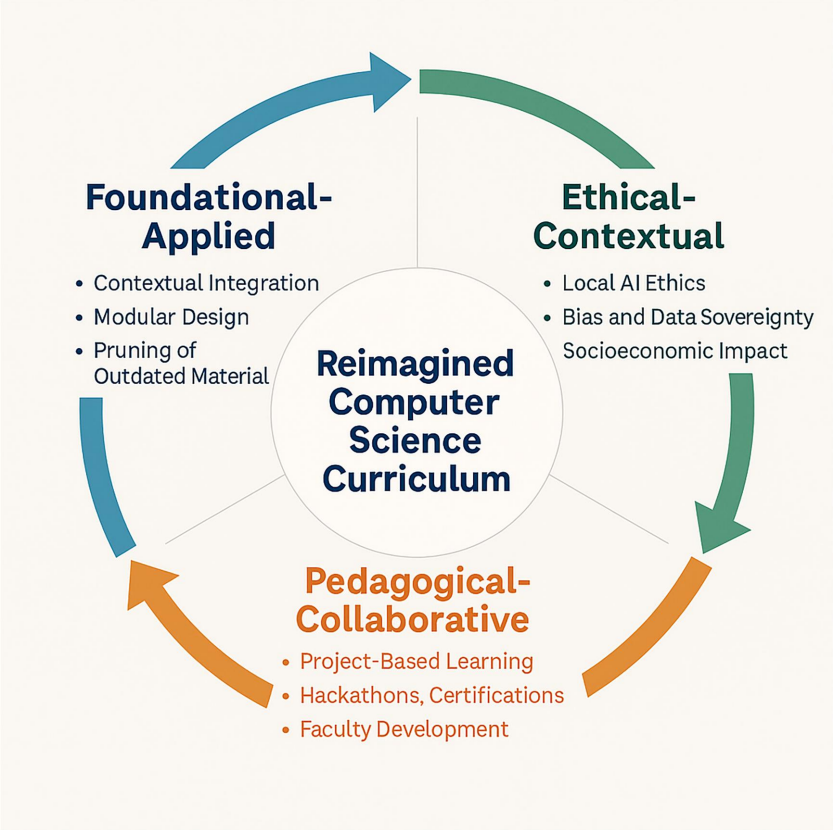


Figure 1. Resource-aware AI literacy curriculum framework (RA-AILCF).

While these dimensions provide the conceptual structure of the framework, their value lies in their practical applicability. To address this, Table 4 translates the proposed framework into actionable curriculum strategies tailored to resource-constrained contexts. While the proposed framework is structured around three core dimensions-foundational-applied integration, ethical-contextual alignment, and pedagogical–collaborative transformation- the operationalisation presented in Table 4 translates these dimensions into a staged, practice-oriented implementation pathway. Each stage reflects a combination of these dimensions, rather than a one-to-one mapping, illustrating how the framework can be enacted progressively within institutional contexts.

The staged approach acknowledges the complexity of curriculum reform in resource-constrained environments, where immediate, large-scale transformation is often impractical. Instead, the framework supports incremental adoption, beginning with foundational adjustments and advancing towards more integrated, collaborative, and context-responsive practices. In this way, the operationalisation provides a pragmatic roadmap that aligns theoretical intent with institutional realities, while maintaining flexibility to accommodate varying levels of readiness, capacity, and resource availability.

Table 4 demonstrates how the framework can be operationalised through concrete curriculum actions, supported by context-sensitive implementation strategies and clearly defined institutional roles. In contrast to models that assume high levels of technological infrastructure, the proposed framework emphasises feasible adoption pathways, including open-source tools, low-bandwidth learning environments, shared institutional resources, and mobile-first platforms.

A key feature of the framework is its emphasis on staged adoption, aligned with DOI principles. Rather than requiring comprehensive curriculum redesign at the outset, institutions can begin by integrating AI components into selected modules, such as programming, data science, or capstone projects and progressively scale these initiatives based on available capacity and observed outcomes. This incremental approach reduces implementation risk while allowing institutions to build internal expertise and infrastructure over time.

Table 4. Operationalisation of the resource-aware AI literacy curriculum framework in resource-constrained contexts.

Framework axis	Core purpose	Curriculum actions	Low-Resource implementation strategies	Primary actors
Foundational-Applied Integration	Preserve core computer science principles while embedding AI literacy and practical competencies.	Integrate AI concepts (e.g. machine learning, data analytics, automation) into existing modules such as programming, algorithms, databases, and systems design; introduce applied AI labs and capstone projects.	Use open-source tools (e.g. Python, TensorFlow, scikit-learn); leverage free cloud platforms (e.g. Google Colab); adopt shared institutional labs; utilise lightweight datasets and offline notebooks where connectivity is limited.	Curriculum designers, lecturers, departmental boards
Ethical-Contextual Alignment	Ensure AI literacy is grounded in local realities, ethical considerations, and societal needs.	Embed ethics modules addressing bias, privacy, and data governance; incorporate African datasets and locally relevant case studies; include multilingual and culturally responsive content.	Use publicly available local datasets; integrate case-based teaching; partner with communities for real-world problem framing; align with national data protection policies.	Lecturers, policy units, institutional ethics committees
Pedagogical-Collaborative Transformation	Shift from theory-dominant instruction to experiential, student-centred, and interdisciplinary learning	Introduce project-based learning, hackathons, interdisciplinary group work, and problem-solving tasks linked to real-world challenges; integrate industry-driven assignments	Implement low-cost project-based assessments; organise virtual guest lectures and mentorship; use mobile-first learning platforms; facilitate peer learning and collaborative projects	Lecturers; students; industry partners
Industry-Academic Collaboration	Strengthen alignment between curriculum and labour market demands	Co-develop course content with industry; embed internships, mentorships, and guest lectures; integrate real-world datasets and tools into teaching	Establish partnerships with local tech firms and startups; use virtual collaboration platforms; engage alumni networks; implement short-term industry immersion programmes	Universities; industry stakeholders; government agencies
Staged Adoption and Diffusion Strategy	Enable gradual and sustainable curriculum transformation aligned with institutional capacity	Pilot AI-integrated modules within selected courses; evaluate outcomes; scale progressively across programmes	Begin with electives or capstone modules; prioritise high-impact courses; adopt modular curriculum updates rather than full redesign; use blended learning to optimise limited resources.	University leadership; curriculum committees
Infrastructure and Resource Adaptation	Address infrastructural constraints while supporting AI-enabled learning	Align curriculum delivery with available technological capacity; prioritise scalable and flexible tools	Use hybrid infrastructure (local servers + cloud access); schedule shared lab usage; prioritise low-bandwidth solutions; adopt open educational resources (OERs)	ICT departments; university management

Furthermore, the framework highlights the importance of multi-stakeholder collaboration. Effective implementation requires coordinated efforts among lecturers, curriculum designers, institutional leadership, industry partners, and policymakers. Industry engagement plays a critical role in ensuring that curricula remain aligned with evolving labour market demands and in providing students with exposure to real-world applications of AI.

Importantly, the framework does not prescribe a rigid curriculum structure but rather provides a flexible, adaptable model that can be tailored to different institutional contexts. Its design recognises the diversity of higher education environments in the Global South and seeks to balance ambition with feasibility.

Overall, the Reimagined Computer Science Curriculum Framework offers a practically grounded and context-sensitive approach to integrating AI literacy into higher education. By linking empirical insights with established theoretical perspectives and operational strategies, it provides a coherent pathway for institutions seeking to navigate the transition from digital literacy to AI literacy in a structured and sustainable manner.

In resource-constrained settings, the framework is intended to be implemented incrementally rather than through wholesale curriculum replacement. A feasible entry strategy is to begin with one or two pilot modules, typically in programming, data science, or capstone courses, in which AI concepts can be

embedded using open-source tools, free cloud credits, shared institutional infrastructure, and locally relevant datasets. Once teaching capacity, student response, and resource feasibility have been evaluated, the model can be scaled across the programme. This staged approach reduces implementation risk while preserving curricular coherence.

7.6. Novel contribution

The framework makes three contributions. First, it moves beyond generic AI literacy competency models by focusing specifically on renewing computer science curricula rather than on general digital education. Second, it is empirically derived from the perspectives of students, lecturers, and ICT managers in Zimbabwean higher education, rather than being proposed solely from conceptual or review-based literature. Third, it incorporates implementation logic for resource-constrained environments by foregrounding staged adoption, open-source ecosystems, hybrid infrastructure, local data relevance, and cross-sector collaboration. It defines the connections between AI literacy's what (foundational and applied content), how (constructivist pedagogy), and why (ethical and contextual purpose). The framework, which combines theoretical and pragmatic factors, opens a sustainable pathway to developing AI-competent graduates who will drive innovation in Zimbabwe's higher education and beyond, thereby rightly situating AI literacy within the African higher education context, which remains underrepresented in global scholarship.

In contrast to previous regional studies, which mainly analyse the obstacles to AI adoption (Al-Zahrani & Alasmari, 2025; Henadirage & Gunarathne, 2025; Maluleke, 2025), the present research identifies a reform process that is shareable and integrates context, tree-cutting, and faculty certification. While most international studies treat AI literacy as technical competence or cognitive understanding, the current inquiry takes the conversation further by elaborating that AI literacy must also encompass ethical grounding, institutional preparedness, and Constructivist pedagogy. The convergence of DOI, TAM, and CLT within a single interpretive framework imparts a new multi-faceted rationale of why AI curriculum reforms are either successful or stalled in the Global South

7.7. Early observations following initial curriculum updating

Although the primary focus of this study was on stakeholder perspectives regarding the transition from digital to AI literacy, the broader triangulated inquiry also indicated that some universities had already begun introducing selected AI-related updates into parts of their Computer Science curricula. The observations presented in this section are therefore not speculative projections, but early implementation-related observations drawn from the wider evidence base considered in the study, including participant accounts and documentary materials reviewed during triangulation. They are presented to illustrate how initial curriculum adjustments are beginning to manifest in practice, while recognising that implementation remains uneven across institutions and is still in its early stages.

The early stages of integrating AI literacy into Computer Science programmes at some Zimbabwean universities have shown observable changes. Although curriculum reform remains ongoing and uneven, some institutions have begun to introduce pilot AI-related modules, revise selected course content, and expand applied learning opportunities. These early observations indicate alignment between curriculum content and industry expectations, with increased student engagement and a stronger emphasis on practical problem-solving in AI-related contexts.

Lecturers who participated in AI literacy workshops and faculty upskilling programmes reported increased confidence in delivering AI-related content and integrating ethical and data-driven perspectives into their pedagogy. In some cases, this growing capacity was associated with the initiation of research projects focused on locally relevant AI applications and ethical considerations.

There were also indications of increased student participation in hackathons and AI-related competitions, which appeared to stimulate the development of solutions targeting local challenges, particularly in sectors such as agriculture and healthcare resource management.

Furthermore, ICT managers overseeing student internships observed that students exposed to revised curricular components appeared more adaptable to AI-oriented work environments. In some instances,

student projects in areas such as climate-smart agriculture, healthcare diagnostics, and local-language processing were featured in national innovation platforms, suggesting early signs of practical application and relevance.

However, implementation remains uneven across institutions. While some universities have initiated curriculum updates, others are still in planning stages, including curriculum review, regulatory approval processes, and staff capacity development. This variation highlights that AI curriculum reform is a gradual, complex process that requires sustained institutional coordination, policy support, and continuous investment in capacity building.

These observations should therefore be interpreted as early indications of implementation rather than comprehensive outcome measures, given that curriculum reform remains ongoing and uneven across institutions.

8. Conclusions, limitations and future works

This research provides crucial information about the shift in focus from digital literacy to AI literacy and the need to redesign the CS curriculum as a subsequent step. The study, through the subtle perceptions and problems of lecturers, industry experts, and students, has brought to light important themes that form part of a comprehensive plan for the future preparation of computing professionals in the AI era. The results of the research emphasise the need for curriculum reform, highlighting the evolving characteristics of AI literacy, the adoption of interdisciplinary methods, the support for lecturer training, the establishment of industry partnerships, the overcoming of institutional barriers, and the development of a comprehensive literacy concept. Stakeholders can orchestrate a more relevant and practical educational landscape and pinpoint the major barriers and facilitators of change in academic institutions. The study's findings serve as a guide for improving CS education, aligning with the needs of an AI-driven world. The findings provide a roadmap for those aiming to upgrade CS education and ensure that it remains relevant in a digitising world. Thus, the study claims that AI will not completely replace CS graduates; however, CS students who are AI-literate, as defined by this study, will replace those who are not. The study further proposed a curriculum framework that reimagined CS, consisting of Foundational-Applied, Ethical-Contextual, and Pedagogical-Collaborative. The innovation of this model lies in the combination of technical skill, ethical localisation, and constructivist teaching, thereby providing a multi-theoretical perspective on curriculum change in African contexts. Rather than positioning AI as a replacement for human expertise, the study advocates for an educational paradigm that prepares graduates to engage critically, ethically, and collaboratively within increasingly AI-mediated professional environments.

9. Limitations

The research is qualitative, so its findings are exploratory and context-specific, which may limit their direct applicability to all computer science departments worldwide or to other education systems. The knowledge obtained is heavily influenced by the views of a certain group of stakeholders. It may not align with the views of all stakeholders. Secondly, although the sample size was suitable for qualitative saturation, it may still limit the generalizability of the findings across different contexts; hence, whilst participant perceptions provide insightful views into systemic trends, they are better utilised as a basis for further comparative, multi-country research rather than for concrete regional generalisation.

10. Future work

The qualitative insights generated by this study provide a basis for several avenues for future research. A quantitative study could utilise surveys to measure the prevalence of the identified perceptions and challenges across a larger, more diverse sample of CS departments and industry organisations. Longitudinal research would be beneficial for observing the implementation of AI literacy initiatives in curricula over time and for determining their effectiveness in preparing graduates. Besides, further qualitative research could examine specific pedagogical strategies for teaching AI ethics or for integrating AI

literacy into non-CS disciplines. Moreover, comparative studies across countries or regions could not only provide insight into how educational policies and industry landscapes shape curriculum reimagining but also shed light on how these factors contribute to the development of AI literacy assessment tools. Lastly, the perspectives collected are indicative of specific experiences. They may not represent the full spectrum of opinions across universities and disciplines. Thus, future research using larger, more diverse samples will play an important role in confirming these findings and deepening the understanding of AI literacy in Zimbabwe's higher education system.

By moving beyond technology-centric curriculum reform towards a resource-aware, contextually grounded, and ethically aligned model of AI literacy, this study provides a scalable pathway for institutions operating under structural constraints.

Author contributions

CRediT: **Belinda Ndlovu**: Conceptualization, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing.

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Data availability statement

The data used for this study is available from the author upon a reasonable time request.

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